

Project Report: E-commerce Return Prediction

1. Problem Statement

High product return rates in e-commerce affect profitability, inventory planning, and customer satisfaction. This project aims to analyze transaction data and predict return likelihood, providing insights to reduce returns. The analysis includes data cleaning, feature engineering, modeling, and interactive visualization using Power BI.

2. Data Overview

The synthetic dataset contains order-level data with fields such as Product_ID, User_ID, Product_Category, Order_Date, Delivery_Date, Product_Price, Order_Quantity, and Discount_Percentage. A Return_Flag column was engineered based on business logic to simulate real-world return patterns.

3. Exploratory Data Analysis (EDA)

Exploratory analysis revealed:

- Overall return rates were around 50%, indicating a balanced target variable.
- Certain product categories exhibited higher return tendencies.
- Delivery delays correlated positively with return likelihood.
- High-value products showed slightly higher return behavior.

4. Machine Learning Models

Several classification models were tested:

- Logistic Regression (baseline model)
- Random Forest Classifier
- XGBoost and LightGBM

Hyperparameter tuning with GridSearchCV was applied, but model performance remained around 50% accuracy, indicating limited predictive power in available features.

5. Model Performance

Model	Accuracy	Precision	Recall	F1 Score
Logistic Regression	50.6%	0.51	0.51	0.51

Random Forest	50.7%	0.51	0.51	0.51
XGBoost	50.8%	0.51	0.51	0.51

6. Observations & Limitations

- Features lacked strong predictive signal for return behavior.
- No class imbalance; target was roughly 50/50.
- Even advanced models (XGBoost, LightGBM) showed minimal improvement.

7. Recommendations

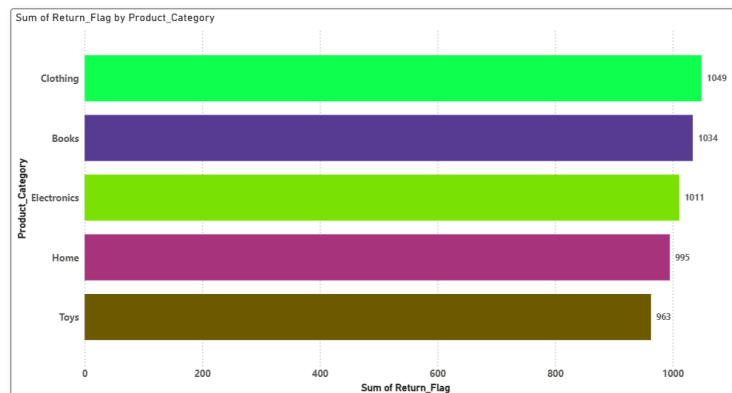
- Collect additional features like user behavior history and product review sentiment.
- Perform text analytics on customer feedback.
- Explore anomaly detection methods to identify atypical returns.

8. Power BI Dashboard Summary

An interactive Power BI dashboard was created to visualize return risk insights. The key visuals include:

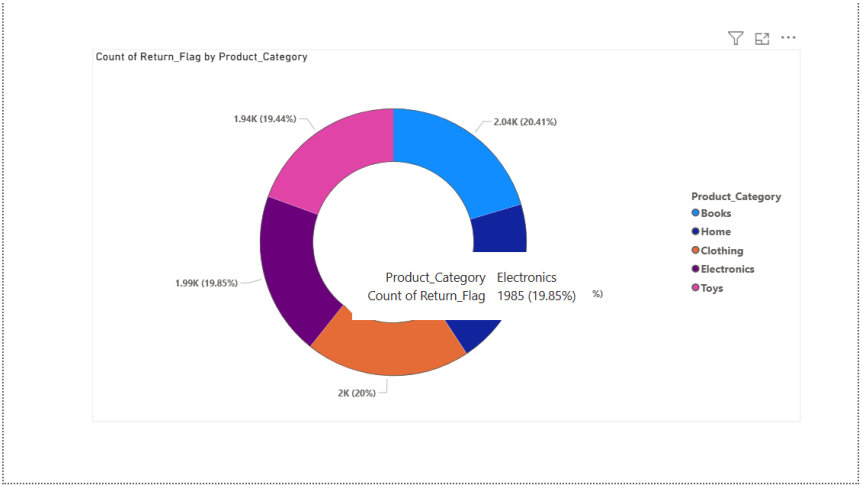
Bar charts

Sum of return flag by product category



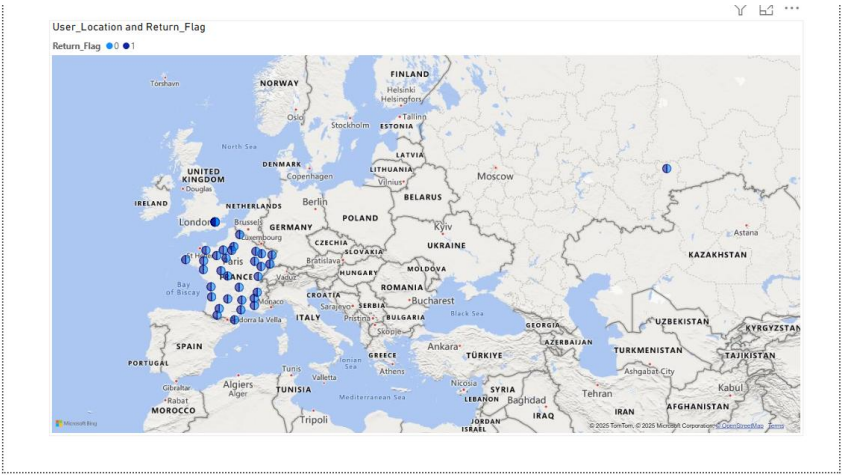
Donut Pie charts

Donut Pie charts of count of return flag by product category



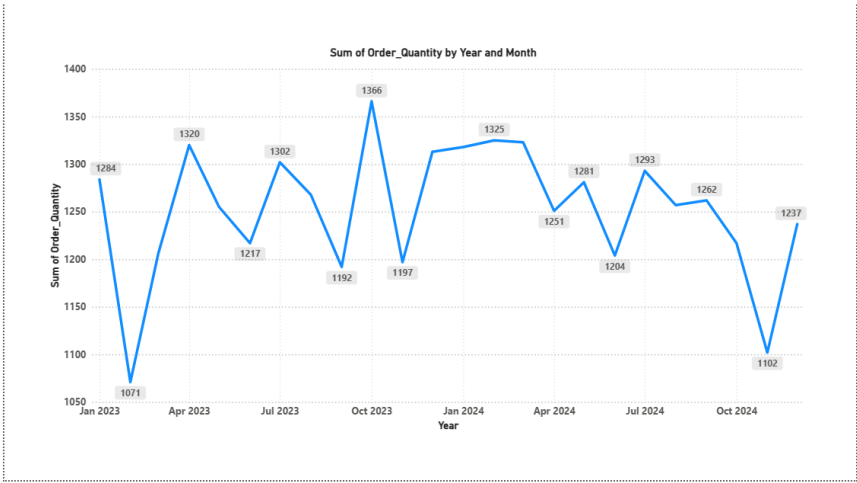
Area chart

User location and there return flag



Line chart

Order Quantity by year and month



Dashboard

Final Dashboard with filter



9. Conclusion

The project demonstrates an end-to-end workflow for predicting and analyzing e-commerce returns. While model accuracy was limited due to synthetic data constraints, the Power BI dashboard provides actionable insights. Future work involves integrating real data, enriching features, and refining predictive models for better performance.